

INITIAL VERTICAL PILE LOAD TEST

Test Report

Pile ID:	TP-01
Report No:	IVPLT-001
Project:	BDD Chawls Redevelopment Project, Worli, Mumbai
Location:	Building-3 Wing B
Client:	Tata Project
Contractor:	NA
Test Date:	09 December 2025

Test conducted as per IS 2911 (Part 4) - 2013

1. Test Results Summary

TEST PASSED ✓

Net Settlement: 4.14 mm (Limit: 12 mm)

TEST LOAD 1050.0 MT (2.5× Design)	MAX SETTLEMENT 7.22 mm	ELASTIC REBOUND 3.08 mm
NET SETTLEMENT 4.14 mm	SAFE LOAD ADOPTED 420.0 MT	DESIGN LOAD 420.0 MT

2. Pile Specifications

Pile ID	TP-01
Pile Diameter	900 mm
Pile Depth	11.51 m
Concrete Grade	M40
Design Safe Load	420 MT
Test Load (2.5× Design)	1050 MT

3. Equipment Details

Hydraulic Jack	As per site specifications
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Ram Area	2551 cm²
Dial Gauge Least Count	0.01 mm
Test Method	IS 2911 (Part 4) - 2013

4. Methodology

The Initial Vertical Pile Load Test was conducted in accordance with IS 2911 (Part 4) - 2013 "Code of Practice for Design and Construction of Pile Foundations - Load Test on Piles".

Test Procedure:

1. Load was applied in increments of 20% of design load up to test load (1050 MT).
2. Each load increment was maintained until rate of settlement was less than 0.1mm/hour.
3. Maximum load was maintained for 24 hours as per IS 2911 requirements.
4. Load was released in the same increments as loading phase.
5. Settlement readings were recorded using four dial gauges at each load increment.

5. Acceptance Criteria (IS 2911 Part 4)

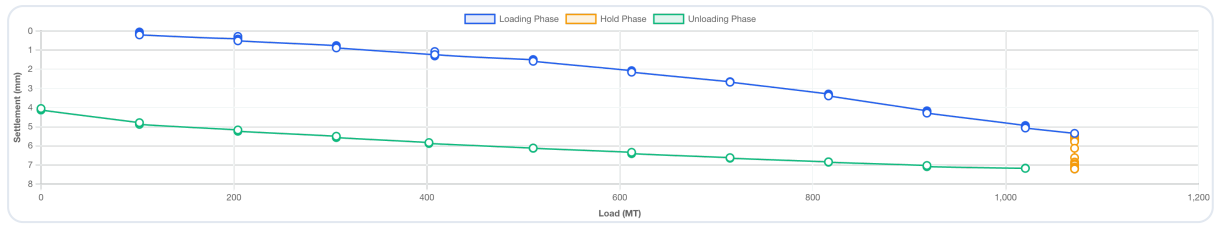
As per IS 2911 (Part 4) - 2013, the acceptance criteria for Initial Vertical Pile Load Test are:

1. Net settlement at design load shall not exceed 12mm or 2% of pile diameter (18.0mm), whichever is less.
2. Safe load shall be taken as two-thirds of the load at which total settlement equals 12mm.
3. Safe load shall not exceed half the load at which total settlement equals 10% of pile diameter (90.0mm).

6. Results Analysis

Maximum Settlement Recorded	7.22 mm
Elastic Rebound	3.08 mm
Net Settlement (Max - Rebound)	4.14 mm
Settlement Limit (IS 2911)	12 mm
Safe Load Adopted	420.0 MT
Governing Criterion	DESIGN
Test Status	PASSED

7. Load vs Settlement Curve



Blue: Loading Phase | Amber: Hold Phase | Green: Unloading Phase

8. Load Test Data

S.NO	PHASE	LOAD (MT)	SETTLEMENT (MM)
1	LOADING	0.00	0.00
2	LOADING	102.04	0.06
3	LOADING	102.04	0.13
4	LOADING	102.04	0.17
5	LOADING	102.04	0.19
6	LOADING	102.04	0.21
7	LOADING	204.08	0.41
8	LOADING	204.08	0.29
9	LOADING	204.08	0.43
10	LOADING	204.08	0.47
11	LOADING	204.08	0.52
12	LOADING	306.12	0.77
13	LOADING	306.12	0.81
14	LOADING	306.12	0.83
15	LOADING	306.12	0.86
16	LOADING	306.12	0.89
17	LOADING	408.16	1.25
18	LOADING	408.16	1.28
19	LOADING	408.16	1.31
20	LOADING	408.16	1.08
21	LOADING	408.16	1.24
22	LOADING	510.20	1.51
23	LOADING	510.20	1.50

S.NO	PHASE	LOAD (MT)	SETTLEMENT (MM)
24	LOADING	510.20	1.53
25	LOADING	510.20	1.56
26	LOADING	510.20	1.59
27	LOADING	612.24	2.08
28	LOADING	612.24	2.11
29	LOADING	612.24	2.13
30	LOADING	612.24	2.13
31	LOADING	612.24	2.16
32	LOADING	714.28	2.65
33	LOADING	714.28	2.65
34	LOADING	714.28	2.67
35	LOADING	714.28	2.68
36	LOADING	714.28	2.69
37	LOADING	816.32	3.30
38	LOADING	816.32	3.33
39	LOADING	816.32	3.35
40	LOADING	816.32	3.38
41	LOADING	816.32	3.40
42	LOADING	918.36	4.18
43	LOADING	918.36	4.20
44	LOADING	918.36	4.23
45	LOADING	918.36	4.27
46	LOADING	918.36	4.30
47	LOADING	1020.40	4.95
48	LOADING	1020.40	5.00
49	LOADING	1020.40	5.03

S.NO	PHASE	LOAD (MT)	SETTLEMENT (MM)
50	LOADING	1020.40	5.06
51	LOADING	1020.40	5.08
52	LOADING	1071.42	5.36
53	HOLD	1071.42	5.40
54	HOLD	1071.42	5.44
55	HOLD	1071.42	5.50
56	HOLD	1071.42	5.53
57	HOLD	1071.42	5.58
58	HOLD	1071.42	5.60
59	HOLD	1071.42	5.62
60	HOLD	1071.42	5.66
61	HOLD	1071.42	5.67
62	HOLD	1071.42	5.68
63	HOLD	1071.42	5.74
64	HOLD	1071.42	5.79
65	HOLD	1071.42	6.14
66	HOLD	1071.42	6.63
67	HOLD	1071.42	6.86
68	HOLD	1071.42	6.95
69	HOLD	1071.42	7.00
70	HOLD	1071.42	7.12
71	HOLD	1071.42	7.17
72	HOLD	1071.42	7.19
73	HOLD	1071.42	7.21
74	HOLD	1071.42	7.22
75	HOLD	1071.42	7.22

S.NO	PHASE	LOAD (MT)	SETTLEMENT (MM)
76	HOLD	1071.42	7.22
77	UNLOADING	1020.40	7.18
78	UNLOADING	1020.40	7.18
79	UNLOADING	1020.40	7.18
80	UNLOADING	918.36	7.08
81	UNLOADING	918.36	7.06
82	UNLOADING	918.36	7.04
83	UNLOADING	816.32	6.87
84	UNLOADING	816.32	6.84
85	UNLOADING	816.32	6.84
86	UNLOADING	714.24	6.66
87	UNLOADING	714.24	6.64
88	UNLOADING	714.24	6.63
89	UNLOADING	612.24	6.41
90	UNLOADING	612.24	6.37
91	UNLOADING	612.24	6.34
92	UNLOADING	510.20	6.14
93	UNLOADING	510.20	6.13
94	UNLOADING	510.20	6.13
95	UNLOADING	402.16	5.89
96	UNLOADING	402.16	5.87
97	UNLOADING	402.16	5.84
98	UNLOADING	306.12	5.58
99	UNLOADING	306.12	5.54
100	UNLOADING	306.12	5.51
101	UNLOADING	204.08	5.25

S.NO	PHASE	LOAD (MT)	SETTLEMENT (MM)
102	UNLOADING	204.08	5.21
103	UNLOADING	204.08	5.18
104	UNLOADING	102.04	4.89
105	UNLOADING	102.04	4.82
106	UNLOADING	102.04	4.80
107	UNLOADING	0.00	4.14
108	UNLOADING	0.00	4.11
109	UNLOADING	0.00	4.06

9. Conclusion

Based on the Initial Vertical Pile Load Test conducted on pile TP-01 at Building-3 Wing B, the following observations and conclusions are drawn:

1. The pile was loaded up to 1050.0 MT (2.5 times the design load of 420 MT) as per IS 2911 (Part 4) - 2013.
2. The maximum settlement recorded at test load was 7.22 mm.
3. After complete unloading, the elastic rebound was 3.08 mm.
4. The net settlement (residual) is 4.14 mm, which is within the permissible limit of 12 mm.
5. The safe load adopted for the pile is 420.0 MT based on the design criterion.
6. **The pile HAS PASSED the Initial Vertical Pile Load Test as per IS 2911 (Part 4) - 2013.**

Therefore, the design safe load of 420 MT can be adopted as the working load capacity for this pile.

Prepared by:

Approved by:

Site Engineer

Project Manager